

Sai Kaushik Mangalampalli

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EDUCATION

Marri Laxman Reddy Institute of Technology and Management

Bachelor of Technology in Computer Science Engineering (Artificial Intelligence & Machine Learning)

Hyderabad, India

Dec 2020 – Aug 2024

- Percentage: 76.8% – First Class with Distinction

TECHNICAL SKILLS

Programming Languages & Web Technologies: Python (Proficient), Java, JavaScript, HTML5, CSS, Tailwind CSS, Bootstrap

Frameworks & Libraries: Flask, FastAPI, Django, React.js

Development Tools & Platforms: Git, GitHub, GitLab, VS Code, Cursor, Antigravity

Database & Cloud Technologies: MySQL, MongoDB, Firebase, Render, Netlify

Development Methodologies: Agile (Scrum), Test-Driven Development (TDD)

WORK EXPERIENCE

Freelance Frontend Developer | Remote

Feb 2025 – Aug 2025

- Delivered responsive UIs for multiple clients using **React.js, HTML5, CSS3, Tailwind CSS, JavaScript**.
- Applied Agile (Scrum) workflows by incorporating client feedback in short iterations and delivering production-ready designs.
- Collaborated with clients and a senior teammate to refine requirements and resolve responsiveness issues across devices.
- Recent work: [Yuga IAS](#) – developed a landing page and UI components with high-resolution desktop optimization (HD, Full HD, 2K, 4K).

PROJECTS

[CodeArena – AI-Powered Multiplayer Coding Platform](#) | [FastAPI](#), [Judge0 API](#), [Gemini API](#), [Render](#), [Netlify](#)

Sep 2025 – Feb 2026

- Built a real-time multiplayer coding platform using Python, FastAPI, and WebSockets supporting 100 concurrent users(up to 5 players per room) with synchronized timers and dynamic leaderboards.
- Integrated Judge0 for sandboxed execution in 4 languages (Python, Java, Go, JS) and Google Gemini for AI-based time-complexity analysis with 50% partial-point scoring for TLE submissions.
- Deployed a secure, high-performance backend with API proxying, CORS restrictions, environment-based secrets, and full CI/CD pipelines; achieved a 250ms average response time under a 500-request load test.
- Resolved critical production-level challenges including WebSocket race conditions (by reconfiguring Gunicorn to a single worker) and in-memory state loss (by implementing a persistent uptime monitor), ensuring 24/7 service reliability.

[Codenames Online – Real-Time Multiplayer Web Game](#) | [React.js](#), [Firebase](#), [Tailwind CSS](#), [JavaScript](#)

May 2025 – Aug 2025

- Architected full-stack real-time multiplayer web application using React.js and Firebase Firestore with complex game state management.
- Built React hooks, real-time chat functionality, automated team assignment logic, and turn-based gameplay mechanics.
- Configured 4-player game mechanics with automatic team balancing, role-based access control, and session persistence.
- Crafted UI with Tailwind CSS featuring interactive game board and lobby system; deployed on Netlify with Firebase security rules.

[Tridle – 3-Word Wordle Variant Game](#) | [React.js](#), [JavaScript](#), [CSS3](#), [Python](#)

Jan 2025 – Mar 2025

- Pioneered a Wordle-based puzzle game where players solve three five-letter puzzles simultaneously using React component architecture.
- Designed a virtual keyboard, color-coded feedback system, and theme toggling using React hooks with smooth CSS transitions.
- Integrated persistent player statistics (games played, wins, win %) using localStorage and implemented a 'Play Again' flow which generates new puzzle sets for replayability.
- Complemented frontend with a Python console version for puzzle generation & well-structured state management for multi-board gameplay.

Other Projects: [Football Arena – Ultimate Goals Challenge](#), [Star Showdown – Higher or Lower Game](#)

CERTIFICATIONS

HackerRank Certifications: Python (Basic), Java (Basic), SQL (Basic), Problem Solving (Basic), Problem Solving (Intermediate), Software Engineer Intern, Software Engineer

FreeCodeCamp Certifications: Machine Learning with Python, Scientific Computing with Python, Data Analysis with Python

CISCO Networking Academy: Programming Essentials in C, Advanced Programming in C

PROGRAMMING PROFILES

LeetCode: [EraOfKaushik003](#) (1815+ problems solved) | **HackerRank:** [saikaushik6637](#) | **Codewars:** [EraOfKaushik003](#) (4 kyu)